

L-2. Reclamation of Disaster Affected River Bed Side and Agriculture Land

 Degraded Soil & Land Improvement	 Medium	 High		 Term
Measure Type	Cost	Preferences by Community	Hazards	Implementation Time of Measures

Descriptions

	Reclamation of agricultural land and riverbed sides damaged by disasters is the process of restoring lands that have been harmed or degraded by human activity or natural disasters (such as floods, landslides, or riverbank erosion). Restoring these lands' ecological function or transforming them into productive for agricultural use are the goals. This procedure is particularly crucial in regions where agricultural or riverbank land has been negatively affected, decreasing its stability, fertility, or usefulness.								
Measures	Reclamation of agricultural land and riverbed sides impacted by catastrophes is essential in the context of Chattogram Hill since the area is vulnerable to natural disasters such droughts, landslides, floods, and river erosion. Heavy rainfall and the region's hilly terrain make the land extremely susceptible to erosion and damage. Reclamation of agricultural land and riverbed sides damaged by disasters necessitates a mix of environmental restoration, engineering solutions, agricultural advances, and community involvement. The sustainability and long-term recovery of this sensitive area will depend heavily on sustainable land management strategies.								
Location	R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
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Applicable Sites	• Flood-Affected Farmlands – Areas degraded by seasonal or flash floods. • Eroded Riverbanks – Lands suffering from sediment loss, instability, and bank collapse. • Landslide-Prone Hillslopes – Degraded agricultural lands requiring stabilization and soil restoration. • Drought-Prone Areas – Requires water conservation techniques for soil moisture retention.								
Slope	Less than 3%		3% to 15%		15% to 30%		30% to 60%		Above 60%
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Design options and performance	• Terracing & Contour Farming – Reduces runoff, improves soil stability, and enhances water retention. • Riverbank Stabilization (Gabions, Vegetative Buffer Zones) – Controls erosion and improves bank strength. • Check Dams & Silt Traps – Slows sediment loss and enhances groundwater recharge.								

	Agroforestry & Reforestation – Combines tree planting with agriculture to restore soil fertility.			
Feasibility criteria	- It is essential to evaluate the physical attributes of the land, including its fertility, texture, and type of soil, as well as the likelihood of catastrophic weather events like droughts and floods. - The preservation of current ecosystems, the restoration of biodiversity, and the prevention of additional environmental deterioration (like soil erosion or forest loss) have to be taken into consideration. It is necessary to ensure that the reclamation methods (such as agroforestry, soil restoration, and vegetation planting) are environmentally sustainable. - The project ensure that it provides local communities long-term financial advantages, such as increased food security, income generating, and job development. Involving the community in the process will ensure their participation and support.			
Operation and maintenance	Reclamation of agricultural land and riverbank sides impacted by disasters requires a number of continuous actions to guarantee the sustainability and long-term success of reclamation projects. These include regular monitoring and evaluation, managing crops and forests, adjusting soil fertility, and adapting to shifting conditions.			
Cost	Medium			
Benefits				
	Type	Low	Medium	High
	Canopy Cover Increase			
	Carbon Sequestration			
	Crop Production Increase			
	Crop Failure Risk Reduction			
	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			
	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			
	Groundwater Recharge			
	Pest and Disease Management			

	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	- Protective embankments, check dams, and vegetation planting are examples of land reclamation techniques that stabilize the soil, lessen erosion, and stop the loss of fertile topsoil. - Reforestation and the planting of natural vegetation are two aspects of land reclamation in the CHT that aid in the restoration of regional ecosystems. - By reducing the land's susceptibility to extreme weather events like floods, droughts, and landslides, the reclamation efforts improve the region's climate change resilience. Enhancing soil management and restoring plant cover are two sustainable land use techniques that help protect the land from these climatic impacts. - The reclamation efforts ensure that the land will continue to be a sustainable and viable resource for upcoming generations in addition to restoring its productive potential.			
Remarks	This measure will aid landscape restoration in CHT of Bangladesh along with creating alternative livelihood opportunities.			